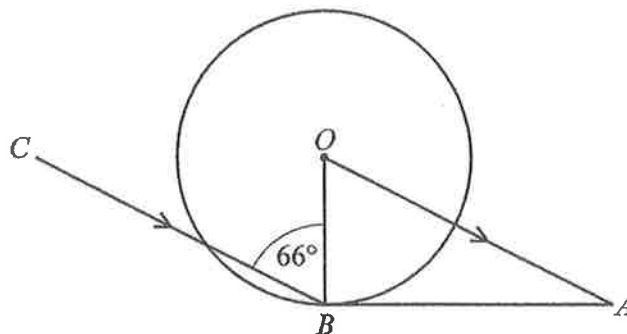


Unit 2.3 Properties of Circles

Paper 1: Basic

1.



AB is a tangent to a circle, centre O .

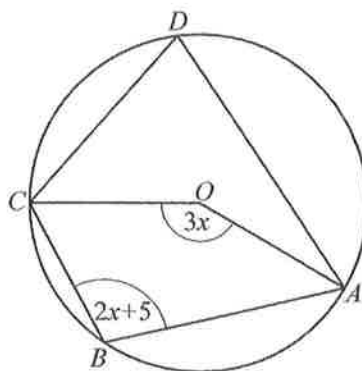
CB is parallel to OA and angle $OBC = 66^\circ$.

Find angle OAB .

[2]

[N20/I/3]

2.



O is the centre of the circle passing through A, B, C and D .

Angle $AOC = 3x^\circ$ and angle $ABC = (2x + 5)^\circ$.

(a) Find, in terms of x , angle ADC .

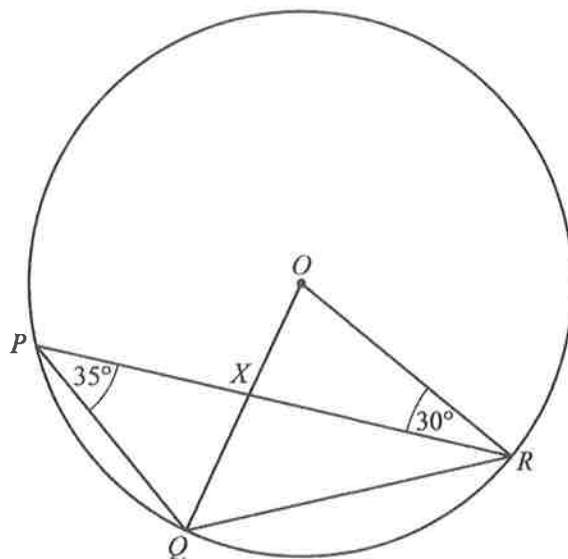
[1]

(b) Find x .

[2]

[N16/I/10]

3.



P , Q and R are three points on a circle, centre O .

Angle $QPR = 35^\circ$ and angle $ORP = 30^\circ$.

OQ and PR cross at the point X .

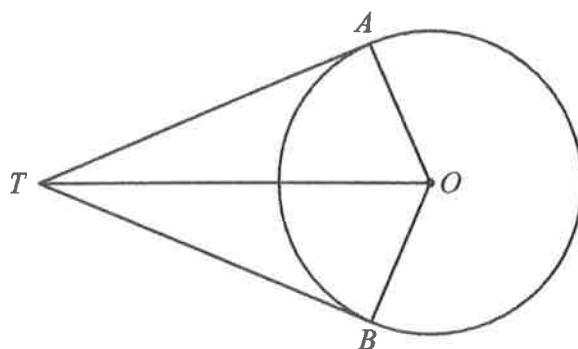
Find angle OQP .

Give reasons for each step of your working.

[4]

[N22/I/17]

4.



(a) TA and TB are tangents to a circle, centre O .

Show that triangle OAT and triangle OBT are congruent.

Give a reason for each statement you make.

[3]

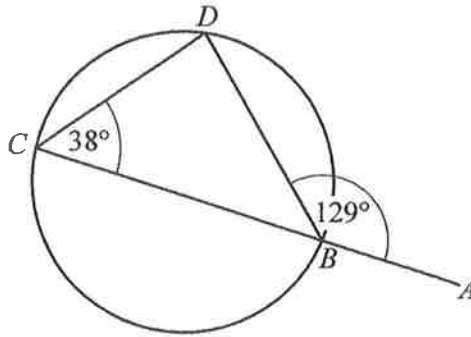
(b) Point P lies on the circumference of the circle so that triangle APT is congruent to triangle BPT .

On the diagram, mark the two possible positions for the point P .

[1]

[N19/I/16]

5.



In the diagram, B , C and D are points on a circle.

ABC is a straight line.

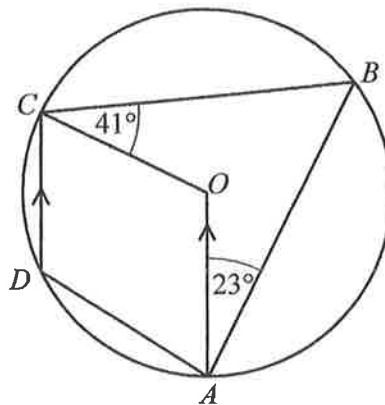
Angle $BCD = 38^\circ$ and angle $ABD = 129^\circ$.

Explain why BC is **not** a diameter of the circle.

[2]

[N18/I/4]

6.



In the diagram, A , B , C and D are four points on a circle, centre O .

OA and CD are parallel.

Angle $OAB = 23^\circ$ and angle $OCB = 41^\circ$.

Find

(a) angle ABC ,

[1]

(b) angle AOC ,

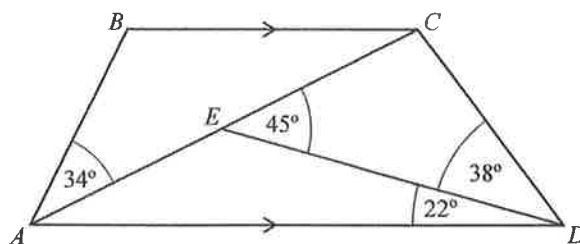
[1]

(c) angle OAD .

[1]

[N18/I/9]

7.



$ABCD$ is a trapezium.

AEC is a straight line.

Angle $BAC = 34^\circ$, angle $DEC = 45^\circ$, angle $CDE = 38^\circ$ and angle $ADE = 22^\circ$.

(a) Calculate the reflex angle ABC .

Give a reason for each step of your working.

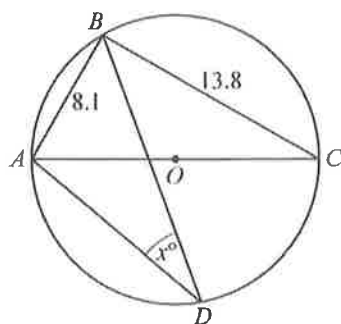
[4]

(b) Explain why a semicircle, with AD as diameter, does not pass through C .

[1]

[N16/I/15]

8.



In the diagram, A , B , C and D are points on a circle, centre O .

$AB = 8.1$ cm, $BC = 13.8$ cm and angle $BDA = x^\circ$.

Find x .

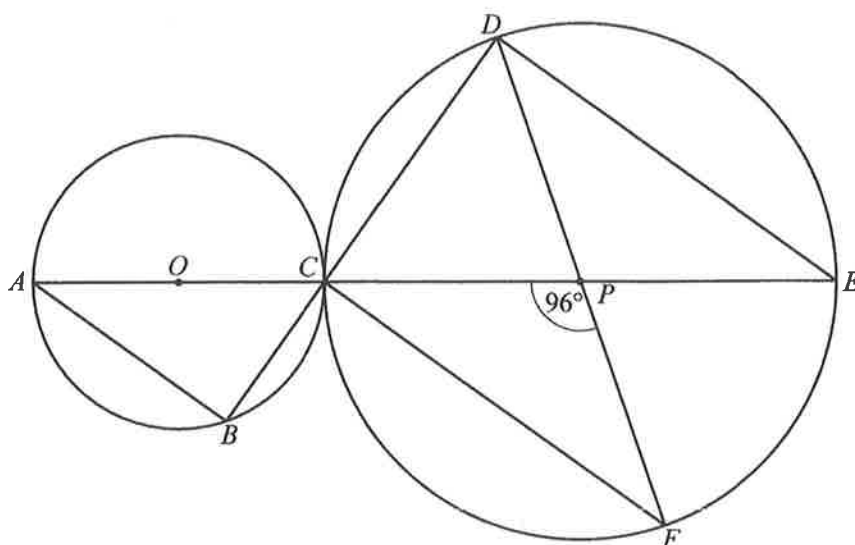
Show your working and give reasons.

[4]

[N17/I/15]

Paper 2: Intermediate

1.



The diagram shows two circles that touch at C .

A , B and C are points on the smaller circle, centre O .

C , D , E and F are points on the larger circle, centre P .

$AOCPE$, BCD and DPF are straight lines.

Angle $CPF = 96^\circ$.

(a) Find angle DEP .

[2]

(b) Show that triangle ABC is similar to triangle FCD .

Give a reason for each statement you make.

[3]

(c) $DE = 7.21$ cm, $DF = 9.70$ cm and $BD = 9.10$ cm and angle $CPF = 96^\circ$.

(i) Calculate AB .

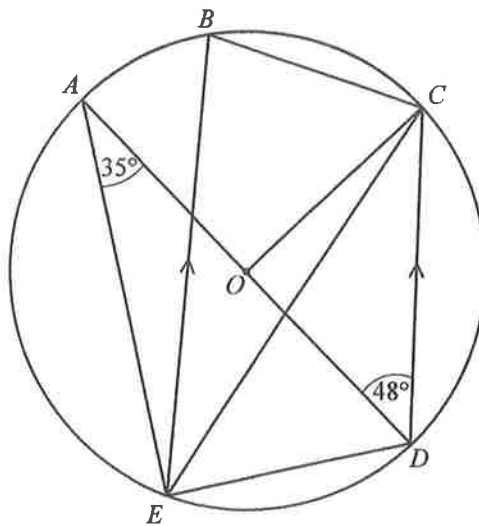
[4]

(ii) Calculate the length of the minor arc AB .

[3]

[N22/II/6]

2.



The diagram shows a circle $ABCDE$, centre O .

BE is parallel to CD .

Angle $EAD = 35^\circ$ and angle $ADC = 48^\circ$.

(a) Find angle BED .

Give a reason for each step of your working.

[3]

(b) Find angle BCE .

Give a reason for each step of your working.

[3]

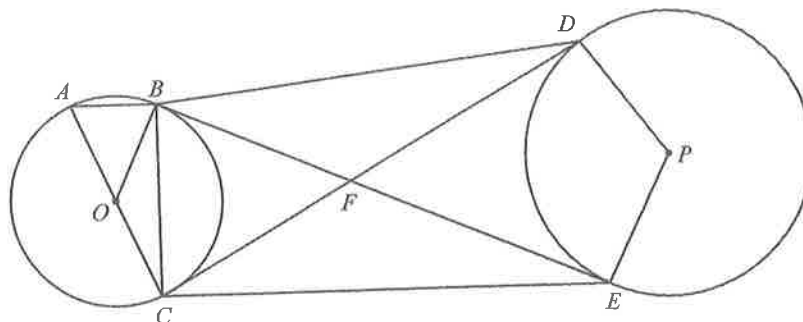
(c) The radius of the circle is 5 cm.

Calculate the area of the major sector $OAEDC$.

[3]

[N21/II/6]

3.



A , B and C are points on the circle with centre O .

D and E are points on the circle with centre P .

BFE and CFD are tangents to both circles.

AOC is a straight line.

(i) Show that triangle BDF is congruent to triangle CEF .

Give a reason for each statement you make.

[3]

(ii) Angle $OAB = x^\circ$.

Find, in terms of x ,

(a) angle OBC ,

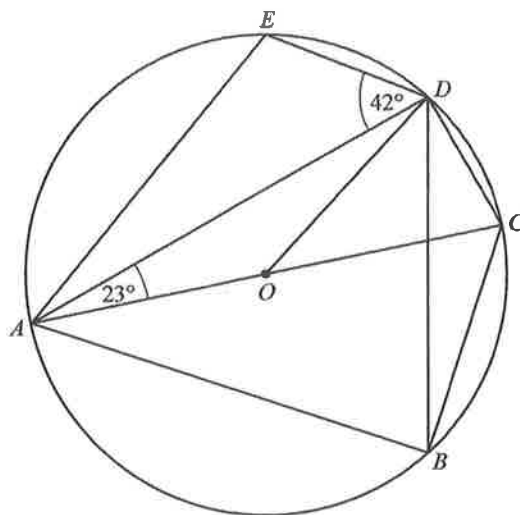
[1]

(b) angle DPE .

[1]

[N20/II/6(a)]

4.



A, B, C, D and E are points on the circle, centre O .

AC is a diameter of the circle.

Angle $OAD = 23^\circ$ and angle $ADE = 42^\circ$.

Find, giving reasons for each answer,

(i) angle ABD ,

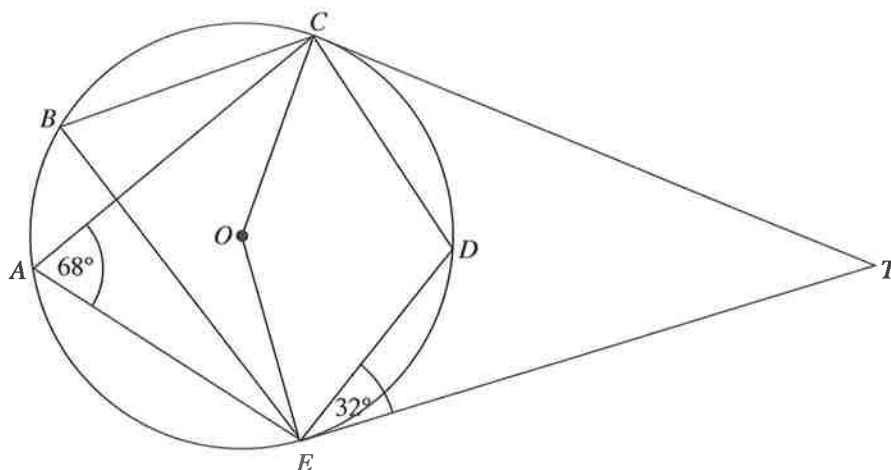
[3]

(ii) angle EAD .

[3]

[N19/II/6(a)]

5. (a)



The diagram shows a circle, $ABCDE$, centre O .

CT and ET are tangents to the circle.

Angle $CAE = 68^\circ$ and angle $DET = 32^\circ$.

Find, giving reasons for each answer,

(i) angle EBC ,

[1]

(ii) angle CDE ,

[1]

(iii) angle CTE ,

[2]

(iv) angle OCD .

[2]

(b) The perimeter of a sector of another circle is 14.8 cm.

The angle of the sector is 1.7 radians.

Calculate the area of the sector.

[3]

[N15/II/5]